

SQL Exercises (HR Database)

SQL JOINS

<https://www.w3resource.com/sql-exercises/joins-hr/index.php>

1. Write a query in SQL to display the first name, last name, department number, and department name for each employee.

```
SELECT E.first_name , E.last_name , E.department_id , D.department_name
FROM employees E
JOIN departments D
ON E.department_id = D.department_id;
```

2. Write a query in SQL to display the first and last name, department, city, and state province for each employee.

```
SELECT E.first_name,E.last_name, D.department_name, L.city, L.state_province
FROM employees E
JOIN departments D
ON E.department_id = D.department_id
JOIN locations L
ON D.location_id = L.location_id;
```

3. Write a query in SQL to display the first name, last name, salary, and job grade for all employees.

```
SELECT E.first_name, E.last_name, E.salary, J.grade_level
FROM employees E
JOIN job_grades J
ON E.salary BETWEEN J.lowest_sal AND J.highest_sal;
```

4. Write a query in SQL to display the first name, last name, department number and department name, for all employees for departments 80 or 40.

```
SELECT E.first_name , E.last_name , E.department_id , D.department_name
FROM employees E
JOIN departments D
ON E.department_id = D.department_id
AND E.department_id IN (80 , 40)
ORDER BY E.last_name;
```

5. Write a query in SQL to display those employees who contain a letter z to their first name and also display their last name, department, city, and state province.

```
SELECT E.first_name,E.last_name, D.department_name, L.city, L.state_province
FROM employees E
JOIN departments D
```

```
ON E.department_id = D.department_id
JOIN locations L
ON D.location_id = L.location_id
WHERE E.first_name LIKE '%Z%';
```

6. Write a query in SQL to display all departments including those where does not have any employee.

```
SELECT E.first_name, E.last_name, D.department_id, D.department_name
FROM employees E
RIGHT OUTER JOIN departments D
ON E.department_id = D.department_id;
```

7. Write a query in SQL to display the first and last name and salary for those employees who earn less than the employee earn whose number is 182.

```
SELECT E.first_name, E.last_name, E.salary
FROM employees E
JOIN employees S
ON E.salary < S.salary
AND S.employee_id = 182;
```

8. Write a query in SQL to display the first name of all employees including the first name of their manager.

```
SELECT E.first_name AS "Employee Name", M.first_name AS "Manager"
FROM employees E
JOIN employees M
ON E.manager_id = M.employee_id;
```

9. Write a query in SQL to display the department name, city, and state province for each department.

```
SELECT D.department_name , L.city , L.state_province
FROM departments D
JOIN locations L
ON D.location_id = L.location_id;
```

10. Write a query in SQL to display the first name, last name, department number and name, for all employees who have or have not any department.

```
SELECT E.first_name, E.last_name, E.department_id, D.department_name
FROM employees E
LEFT OUTER JOIN departments D
ON E.department_id = D.department_id;
```

11. Write a query in SQL to display the first name of all employees and the first name of their manager including those who does not working under any manager.

```
SELECT E.first_name AS "Employee Name",
M.first_name AS "Manager"
```

```
FROM employees E
LEFT OUTER JOIN employees M
ON E.manager_id = M.employee_id;
```

12. Write a query in SQL to display the first name, last name, and department number for those employees who works in the same department as the employee who holds the last name as Taylor.

```
SELECT E.first_name, E.last_name, E.department_id

FROM employees E
JOIN employees S
ON E.department_id = S.department_id
AND S.last_name = 'Taylor';
```

13. Write a query in SQL to display the job title, department name, full name (first and last name) of employee, and starting date for all the jobs which started on or after 1st January, 1993 and ending with on or before 31 August, 1997.

```
SELECT job_title, department_name, first_name || ' ' || last_name AS Employee_name, start_date
FROM job_history
JOIN jobs USING (job_id)
JOIN departments USING (department_id)
JOIN employees USING (employee_id)
WHERE start_date >= '1993-01-01' AND start_date <= '1997-08-31';
```

14. Write a query in SQL to display job title, full name (first and last name) of employee, and the difference between maximum salary for the job and salary of the employee.

```
SELECT job_title, first_name || ' ' || last_name AS Employee_name,
       max_salary-salary AS salary_difference
FROM employees
NATURAL JOIN jobs;
```

15. Write a query in SQL to display the name of the department, average salary and number of employees working in that department who got commission.

```
SELECT department_name, AVG(salary), COUNT(commission_pct)
FROM departments
JOIN employees USING (department_id)
GROUP BY department_name;
```

16. Write a query in SQL to display the full name (first and last name) of employees, job title and the salary differences to their own job for those employees who is working in the department ID 80.

```
SELECT job_title, first_name || ' ' || last_name AS Employee_name,
       max_salary-salary AS salary_difference
FROM employees
NATURAL JOIN jobs
WHERE department_id = 80;
```

17. Write a query in SQL to display the name of the country, city, and the departments which are running there.

```
SELECT country_name,city, department_name
```

```
FROM countries
```

```
JOIN locations USING (country_id)
```

```
JOIN departments USING (location_id);
```

18. Write a query in SQL to display department name and the full name (first and last name) of the manager.

```
SELECT department_name, first_name || ' ' || last_name AS name_of_manager
```

```
FROM departments D
```

```
JOIN employees E
```

```
ON (D.manager_id=E.employee_id);
```

19. Write a query in SQL to display job title and average salary of employees.

```
SELECT job_title, AVG(salary)
```

```
FROM employees
```

```
NATURAL JOIN jobs
```

```
GROUP BY job_title;
```

20. Write a query in SQL to display the details of jobs which was done by any of the employees who is presently earning a salary on and above 12000.

```
SELECT a.*
```

```
FROM job_history a
```

```
JOIN employees m
```

```
ON (a.employee_id = m.employee_id)
```

```
WHERE salary >= 12000;
```

21. Write a query in SQL to display the country name, city, and number of those departments where at least 2 employees are working.

```
SELECT country_name,city, COUNT(department_id)
```

```
FROM countries
```

```
JOIN locations USING (country_id)
```

```
JOIN departments USING (location_id)
```

```
WHERE department_id IN
```

```
(SELECT department_id
```

```
FROM employees
```

```
GROUP BY department_id
```

```
HAVING COUNT(department_id)>=2)
```

```
GROUP BY country_name,city;
```

22. Write a query in SQL to display the department name, full name (first and last name) of manager, and their city.

```
SELECT department_name, first_name || ' ' || last_name AS name_of_manager, city
```

```
FROM departments D
```

```

JOIN employees E
ON (D.manager_id=E.employee_id)
JOIN locations L USING (location_id);

```

23. Write a query in SQL to display the employee ID, job name, number of days worked in for all those jobs in department 80.

```

SELECT employee_id, job_title, end_date-start_date DAYS
FROM job_history
NATURAL JOIN jobs
WHERE department_id=80;

```

24. Write a query in SQL to display the full name (first and last name), and salary of those employees who working in any department located in London.

```

SELECT first_name || ' ' || last_name AS Employee_name, salary
FROM employees
JOIN departments USING (department_id)
JOIN locations USING (location_id)
WHERE city = 'London';

```

25. Write a query in SQL to display full name(first and last name), job title, starting and ending date of last jobs for those employees with worked without a commission percentage.

```

SELECT CONCAT(e.first_name, ' ', e.last_name) AS Employee_name, j.job_title, h.*
FROM employees e
JOIN
    (SELECT MAX(start_date), MAX(end_date), employee_id
     FROM job_history
     GROUP BY employee_id) h ON e.employee_id=h.employee_id
JOIN jobs j ON j.job_id=e.job_id
WHERE e.commission_pct = 0;

```

26. Write a query in SQL to display the department name and number of employees in each of the department.

```

SELECT d.department_name, e.*
FROM departments d
JOIN
    (SELECT count(employee_id), department_id
     FROM employees
     GROUP BY department_id) e USING (department_id);

```

27. Write a query in SQL to display the full name (first and last name) of employee with ID and name of the country presently where (s)he is working.

```

SELECT first_name || ' ' || last_name AS Employee_name, employee_id, country_name

```

```
FROM employees  
JOIN departments USING(department_id)  
JOIN locations USING( location_id)  
JOIN countries USING( country_id);
```